

A Mini Project Report
on
LUNG CANCER DETECTION USING XCEPTION LEARNING

BACHELOR OF ENGINEERING
in
ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

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2025-2026

**Methodist College of Engineering and Technology,
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DECLARATION BY THE CANDIDATES

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CERTIFICATE BY THE SUPERVISOR

This is to certify that this Mini Project work entitled “**LUNG CANCER DETECTION USING XCEPTION LEARNING**” by **Mohammed Inabah Khan (160723748009)**, **Bannenola Akhil (160723748018)** and **Guttula Yeshwanth (160723748041)** submitted in partial fulfilment of the requirements for the degree of Bachelor of Engineering in Department of Artificial Intelligence and Machine Learning, during the academic year 2025-2026, is a Bonafide record of work carried out by them.

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CERTIFICATE BY HEAD OF THE DEPARTMENT

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PROJECT APPROVAL CERTIFICATE

This is to certify that this Mini Project work entitled **“LUNG CANCER DETECTION USING XCEPTION LEARNING”** by **Mohammed Inabah Khan (160723748009), Bannenola Akhil (160723748018) and Guttula Yeshwanth (160723748041)** submitted in partial fulfilment of the requirements for the degree of Bachelor of Engineering in Department of Artificial Intelligence and Machine Learning during the academic year 2025-2026, is a bonafide record of work carried out by them.

INTERNAL

EXTERNAL

HOD

ACKNOWLEDGEMENT

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Vision & Mission

VISION

To become a leader in providing Computer Science & Engineering education with emphasis on knowledge and innovation.

MISSION

- M1:** To offer flexible programs of study with collaborations to suit industry needs
- M2:** To provide quality education and training through novel pedagogical practices
- M3:** To Expedite high performance of excellence in teaching, research and innovations.
- M4:** To impart moral, ethical valued education with social responsibility.

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Graduates of Compute Science and Engineering at Methodist College of Engineering and Technology will be able to:

- PEO1:** Apply technical concepts, Analyze, synthesize data to Design and create novel products and solutions for the real-life problems.
- PEO2:** Apply the knowledge of Computer Science Engineering to pursue higher education with due consideration to environment and society.
- PEO3:** Promote collaborative learning and spirit of team work through multidisciplinary projects
- PEO4:** Engage in life-long learning and develop entrepreneurial skills.

Program Specific Outcomes

At the end of 4 years, Compute Science and Engineering graduates at MCET will be able to:

- PSO1:** Apply the knowledge of Computer Science and Engineering in various domains like networking and data mining to manage projects in multidisciplinary environments.
- PSO2:** Develop software applications with open-ended programming environments.
- PSO3:** Design and develop solutions by following standard software engineering principles and implement by using suitable programming languages and platforms

PROGRAM OUTCOMES

At the end of the four-year Bachelor of Engineering Program, engineering graduates will be able to:

PO1. Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

ABSTRACT

Lung cancer is one of the leading causes of cancer-related deaths worldwide. Early and accurate diagnosis is critical for improving patient survival rates. However, manual analysis of chest CT scan images by radiologists is time-consuming and may be subject to human error and variability. With the increasing availability of medical imaging data, there is a growing need for automated systems that can assist in the detection and classification of lung cancer.

This project proposes a Deep Learning-based Lung Cancer Detection System using Convolutional Neural Networks (CNN) with Xception transfer learning. The system classifies CT scan images into three categories: Benign Cases, Malignant Cases, and Normal Cases using the IQ-OTHNCCD Lung Cancer Dataset. Image preprocessing and data augmentation techniques such as rotation, zooming, and horizontal flipping are applied to improve model generalization and classification performance. The pre-trained Xception model is used as the feature extraction backbone, while custom classification layers are added for multi-class prediction. The model is trained and optimized using transfer learning techniques to achieve accurate classification results.

A Streamlit-based web application is developed to provide an interactive interface for uploading CT scan images and obtaining prediction results along with confidence scores. By integrating transfer learning with medical image classification, the system enhances diagnostic support capabilities, reduces manual workload, and demonstrates the potential of artificial intelligence in medical image analysis and clinical decision support systems.

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